The Effect of Air Temperature on the Lift Produced by a NACA 2412 Airfoil

I. Introduction

Lift is generated when an airfoil generates a pressure difference at a positive angle of attack, high pressure forms beneath the airfoil, and the low pressure develops above it,producing an upward force(nasa.com). At a negative angle of attack this proportion is opposite, high pressure on the top and low pressure on the bottom, producing a downward force. Lift can be described using Bernoulli's principle in combination with Newton's laws, as the airfoil imparts downward momentum to the surrounding air.

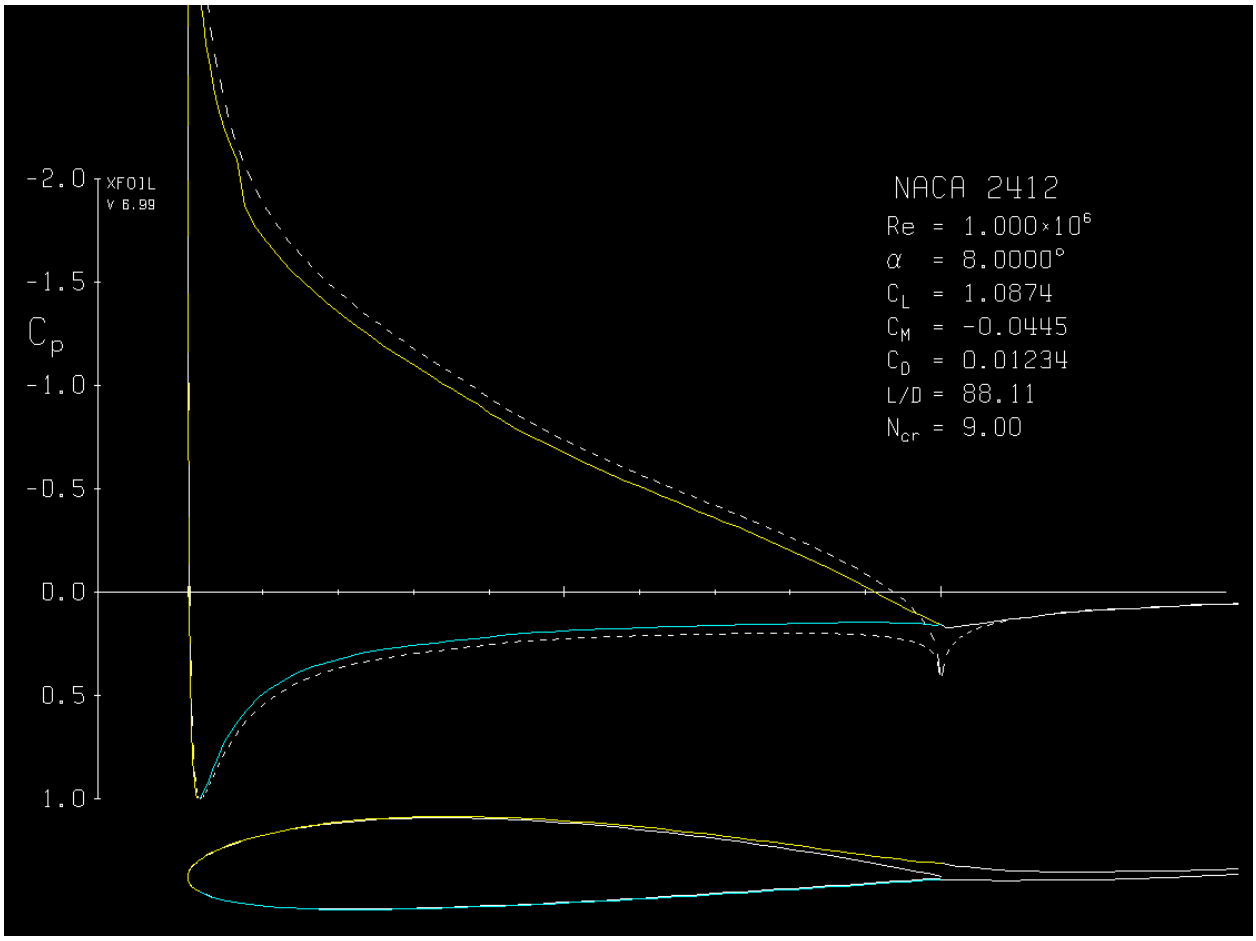
The magnitude of the lift produced by an airfoil depends on several variables, including air density, airflow velocity, angle of attack, and airfoil geometry. In this investigation I will be using a constant testing apparatus, the angle of attack fixed at 8 degrees, which lies within the linear lift region of the NACA 2412 airfoil, which means basically the airfoil actually acts like an airfoil at these conditions and no stalling occurs. The amount of lift an airfoil creates can be expressed by this formula:

$L = \frac{1}{2}\rho\upsilon^2 S C_L$ (Nasa.gov)

Where L is lift, ρ is air density, υ is velocity airflow, S is the airfoil surface area, and C_L

The coefficient of lift is not something I can measure with the tools that I have, therefore

I will be using a very well experimented airfoil, the NACA 2412 (airfoiltools.com). Using MIT's software; Xfoil, I found the coefficient of lift for the specific angle of attack and Reynolds number used in this investigation was determined to be 1.0874, this was assumed constant under the controlled conditions.



To correlate temperature with density mathematically we will use the density equation (in the databooklet):

$$\rho = \frac{m}{v}$$

Where *m i*s mass and *v* is volume, to relate temperature we shall use a modified version of the ideal gas law:

$$PV = \frac{m}{M}RT = \rho = \frac{m}{V} = \frac{PM}{RT}$$

With the final equation being $\rho = \frac{P}{RT}$

Such that: P is absolute pressure of the air measured in pascals, the altitude where I will be conducting test (225m) absolute pressure, P=98.7 KPa, T being temperature in Kelvin, and R being the specific gas constant for air which is 287J(k$g^{-1} \bullet K^{-1}$)

Interpreting this equation logically we can see that when temperature increases, density decreases, and when temperature decreases density increases. This directly affects the lift produced by the airfoil, because lift is proportional to air density in the lift equation.

Since air density changes with temperature, I am motivated to investigate whether air temperature influences the lift produced by an airfoil. My passion for aerospace engineering fuels my interest in this research, as I want to determine if varying air temperature causes measurable differences in the amount of lift created by this airfoil.

II. Research Question

Does the temperature of the air interacting with a NACA 2412 airfoil at a constant airflow velocity of 21.5 m/s affect the amount of lift generated ?

III. Hypothesis and Variables

I hypothesize that the air temperature will affect the lift produced by the airfoil. I predict that cooler air will produce greater lift than hotter air, because colder air is denser than warmer air, increasing the air density term in the lift equation and therefore increasing lift.

Independent Variable: Air temperature.

Dependent Variable: Lift produced by the airfoil.

Controlled Variables: Same airfoil design, Same wind tunnel and airflow settings, same sensor placement,atmospheric pressure, duration of each trial and angle of attack.

Materials and Equipment

1. 1x Pitsco Airtech 40 wind tunnel
2. 1x pressure sensor- Vernier go direct force and acceleration sensor
3. 1x temperature sensor - Vernier go direct temperature sensor
4. 1x custom designed airfoil - 3D printed NACA 2412 airfoil
5. 1x computer with the Vernier data logging software link-(https://www.vernier.com/downloads/graphical-analysis/?srsltid=AfmBOoregZq03jzdQ6JnULKfzqSk1vDmwxcwSU9pWCOHUVN6YG6oiDkn)
6. Access to outdoor air (cold air conditions)
7. Access to indoor room temperatures
8. Access to electricity

v. *Procedure*

- Insert the custom NACA 2412 airfoil into the wind tunnel mounting slot making sure it is secure
- Attach the Vernier force sensor securely to the 3D printed mounting structure holding the airfoil.
- Place the temperature sensor inside of the wind tunnel.
- Connect both the four sensor and the temperature sensor to the computer verifying proper data collection using the veneer graphical analysis software

For the room temperature air trials

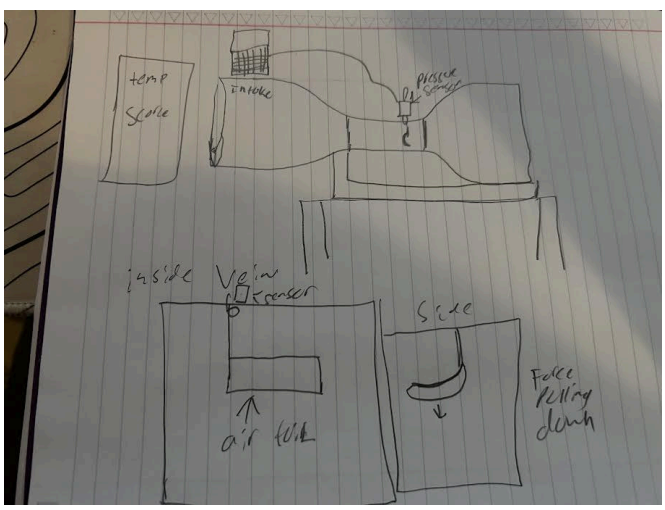
- Place the setup indoors at room temperature.

- Run the tunnel for 5 seconds, then turn it off.
- Record the pressure data that is given to you at the 5 Second mark.
- Repeat this process **six times** for reliability.

Cold air trials

- Move the wind tunnel set up indoors to room temperature environment and allow the air inside the tunnel to equilibrate with the indoor air (wait 30).
- Record the air temperature while the wind tunnel is running using the temperature sensor.
- Run the wind tunnel for 5 seconds and record the lift created at exactly 5 seconds.
- Repeat this step six times.

Store all data in the computer or whatever tracking software you are using for later analysis.



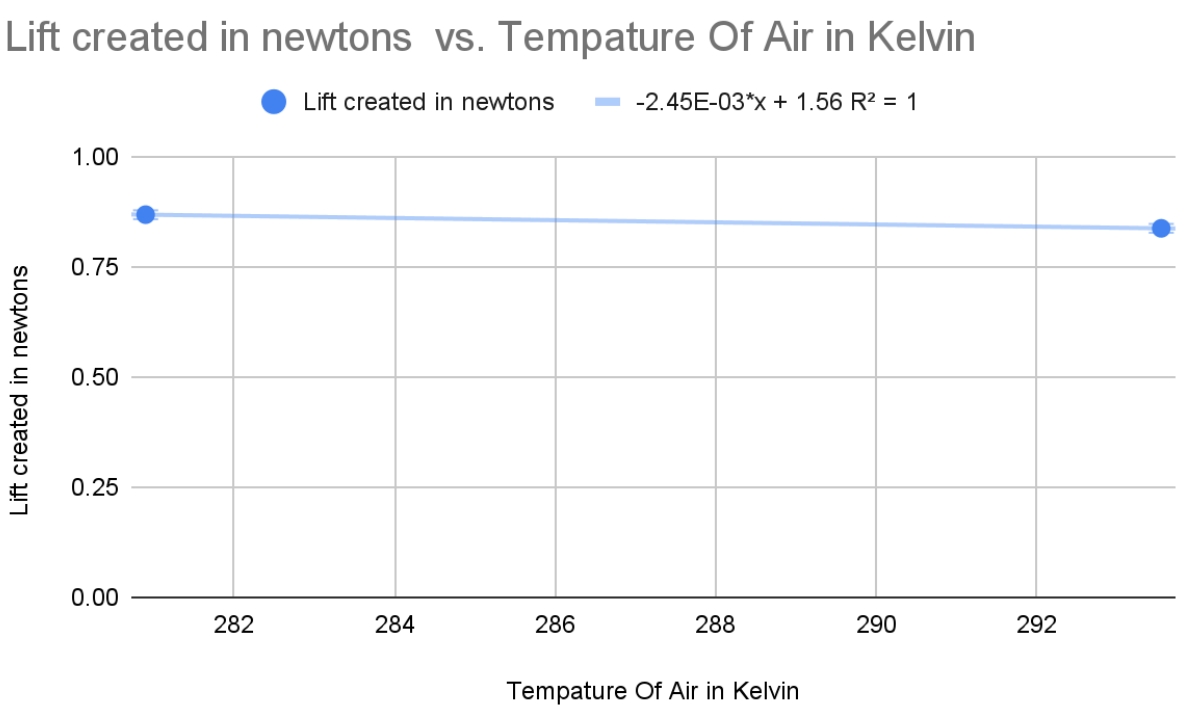
Safety and accuracy

All appropriate safety precautions were followed, including staying a safe distance from moving parts of the wind tunnel and ensuring all electrical connections were secure.

To improve accuracy and reliability, all trials were conducted using the same wind tunnel settings, airfoil orientation, and sensor placement. The angle of attack remained constant throughout the investigation, ensuring that the temperature was the only intentionally varied variable. Multiple trials were performed for each temperature condition, and precise force versus time data was recorded using digital sensors to minimize human error.

Data Analysis

Trial#	1	2	3	4	5	6	7	8	9	10	11	12
Temperature in Kelvin	280.9 $\pm.05$	280.9 $\pm.05$	280.9 $\pm.05$	280.9 $\pm.05$	280.9 $\pm.05$	280.9 $\pm.05$	293.5 $\pm.05$	293.5 $\pm.05$	293.5 $\pm.05$	293.5 $\pm.05$	293.5 $\pm.05$	293.5 $\pm.05$
Lift Created In newtons	.86	.89	.86	.87	.87	.88	.84	.83	.84	.82	.84	.85



The graph of **lift vs temperature** shows that the lift produced by the NACA 2412 airfoil decreased as air temperature increased. In the cold air condition (290.9 k), the mean lift was

0.868 N with an uncertainty of 0.002 N, while in the warm air conditions (293.5 k), the mean lift decreased to 0.837 N with an uncertainty of 0.002 N. This data shows a decrease of approximately 0.031 N, this data is consistent with the lift equations that we have derived that correlates temperature with lift, as colder denser equates to more lift than warmer less dense air. Using six trials for each condition allows us to find a mean and standard deviation, this shows that this experiment is repeatable, the measurement of uncertainty will be propagated using the uncertainty equation from the Ib data booklet.

$$\frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$$

The equation for the uncertainties in the lift equation will be.

$$\frac{\Delta L}{L} = \frac{\Delta \rho}{\rho} + 2\frac{\Delta v}{v} + \frac{\Delta S}{S} + \frac{\Delta C_L}{C_L}$$

The coefficient of lift uncertainty is assumed to be negligible because the Xfoil software reported a constant coefficient of lift for the airfoil(Xfoil).

We can assume that the average uncertainty in 3d printing is 0.5% (Formlabs.com)

Uncertainty in lift = $\frac{.05}{293.5} + \frac{.05}{21.5} + \frac{0.0004}{0.00572}$

This gives me an uncertainty of 0.00249 which is too small to show up on the graph.

The Vernier force sensor measures lift in only the vertical direction not the horizontal direction, minimizing the influence of horizontal forces such as **drag** on the airfoil. These results support the hypothesis that colder, denser air produces greater lift using a constant airfoil.

Conclusion

The outcome of this investigation shows that air temperature does in fact affect the lift produced by the NACA 2412 airfoil at a constant airflow velocity of 21.5 m/s and an angle of attack of 8

degrees. The average lift in the cold air conditions was 0.868 $\pm$ 0.002 N, while the average lift for the warm air conditions was 0.837 $\pm$ 0.002 N. This is a decrease of 0.031 N, this demonstrates an inverse relationship between the air temperature and lift produced. This aligns with the theoretical equation that at lower temperatures produces high air density, which in turn increases lift according to the lift equation.

$L = \frac{1}{2}\rho\upsilon^2 S C_L$ (Nasa.gov)

The uncertainties in the measurements were considered by conducting six trials per condition and calculating standard deviations, demonstrating that the observed difference in lift is statistically meaningful. Furthermore, the force sensor measured only the vertical component of force, ensuring that horizontal forces such as drag did not influence the recorded lift values.

These results align with scientific theory, which states that lift is proportional to air density and that air density changes inversely with fluctuations in temperature at a constant pressure. Therefore, the hypothesis that colder air produces greater lift than warmer air is supported. This investigation shows that even small changes in air temperature or density can cause real changes in lift, which is very important in the field of aerospace.

Evaluation

While this investigation shows that air temperature affects the lift produced by a NACA 2412 airfoil, it is important to point out some limitations and inconsistencies that may have occurred. One of these is the relatively small difference in temperature of about 13 degrees K, this may have reduced the magnitude of the observed effect, this small change in temperature would make

the measurement uncertainty increase. Another limitation may be airflow in the wind tunnel, the airflow may have not been perfectly uniform or constant, I was only able to measure the speed of the wind with an anemometer which did in fact stay constant over about a 30 second interval, but this does not account for the air being completely uniform. At last the coefficient of lift was determined to remain constant, this may not be entirely true, as small variations in the actual design and manufacturing of the airfoil or a change in Reynolds number.

The impact that these limitations have on the investigation is likely small given that multiple trials were conducted. But this could reduce the accuracy of the investigation. To improve the investigation I could have used more sensitive or calibrated sensors, along with having a pressure sensor for the absolute pressure for the day of the test, another could be using sensors to verify that air is uniformly moving throughout the wind tunnel. All of these things would greatly reduce the measurement uncertainties. Finally I could have tested multiple airfoils at different angles, this would strengthen the reliability of the finding that temperature does in fact affect the lift that an airfoil creates.